

Activity: Implement Object Detection with YOLO

Introduction

In this activity, you'll implement **object detection** using the **YOLOv8 model** from the Ultralytics library, a powerful tool for real-time object detection in images. You'll apply a pre-trained YOLOv8 model to detect objects in various scenarios (e.g., street, home, traffic), experiment with different confidence thresholds, and analyze how these thresholds affect detection performance and processing speed.

Learning Outcomes

By the end of this activity, you will:

- Use a pre-trained YOLO model to detect objects in images.
- Experiment with different confidence thresholds and analyze how YOLO balances speed vs. accuracy.

How to Run Your `.ipynb` File on Colab (Recommended)

Option 1: Upload Directly

1. Go to <https://colab.research.google.com>
2. Click **"Upload"** tab → Select your `.ipynb` file from your computer
3. Colab will open it in a new tab
4. Make sure to go to `Runtime → Change runtime type` and set:
 - **Hardware accelerator:** `GPU`

Option 2: Mount from Google Drive

If your notebook is stored in Drive:

1. Open Colab
2. Choose **"Google Drive"** tab
3. Navigate to the `.ipynb` file and open it

Before Running Any Cell

Add a **setup cell at the top** of your notebook to ensure dependencies are installed:

```
!pip install -q torch torchvision transformers datasets matplotlib numpy timm datasets
```

Then Run Cell-by-Cell

1. Start with installation
2. Set up device check and conditional DataLoader
3. Proceed through data, model, training, and evaluation

Optional: Save Outputs Back to Google Drive

If you want to save trained model or results:

```
from google.colab import drive
drive.mount('/content/drive')

# Example to save
torch.save(model_vit.state_dict(), '/content/drive/MyDrive/vit_cifar10.pth')
```

Otherwise if you are using your machine follow the following Steps:

Working with Dev Container

To complete this assignment in a reliable and fully configured environment, please refer to the instructions in the file: `README-devcontainer.md`. This guide walks you through opening the assignment in **Visual Studio Code** using a **Dev Container**, which automatically installs all necessary Python libraries and tools. Following that setup ensures that the notebook runs smoothly without manual configuration or missing dependencies. Make sure to open the **main activity folder** in VS Code and follow the steps outlined in the Dev Container guide before starting the notebook.

Starter File

- Open the notebook: `activity-implement-object-detection-with-YOLO.ipynb` in the `Code` folder.
- Run it locally or in Google Colab.
- Required libraries: `ultralytics`, `opencv-python`, `numpy`, `matplotlib`, `opendatasets`.
- Use the provided image in images under name of `img1.png`, `img2.png`, `img3.png`, `img4.png`

Activity Code Steps

This is a guided learning activity. You'll:

1. Install and Import Required Libraries

Install libraries (`opencv-python`, `numpy`, `matplotlib`, `ultralytics`, `opendatasets`) and import them, including `YOLO` from Ultralytics, to set up the environment for object detection.

2. Load the Pre-trained YOLOv8 Model

3. Process Images and Perform Object Detection

4. Experiment with Confidence Thresholds

Run object detection with varying confidence thresholds (e.g., 0.1, 0.4, 0.7), record the number of detections and processing time (e.g., 4.2ms for one run)

Conclusion

In this activity, you:

- Loaded and applied a pre-trained YOLOv8 model for object detection.
- Processed images from various environments (street, home, traffic) using YOLO.
- Experimented with different confidence thresholds, observing their impact on detection results.
- Measured and compared the number of detections (e.g., 7 persons) and processing speed (e.g., 4.2ms).